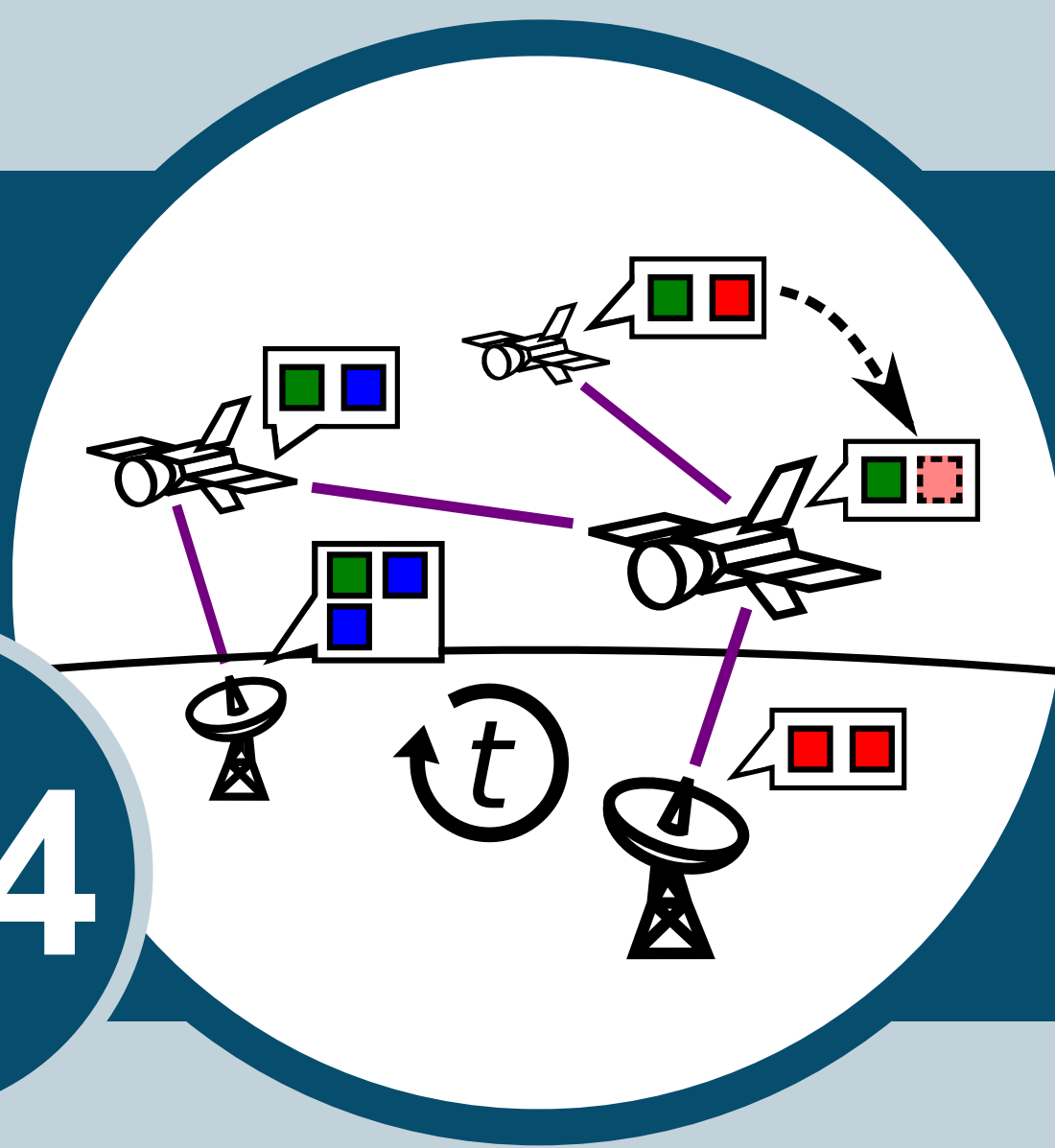


Models and Methods for a Self-Organising Distributed Computing Platform for Satellite Constellations

Ayça Kula Arslan, Björn Annighöfer

C04



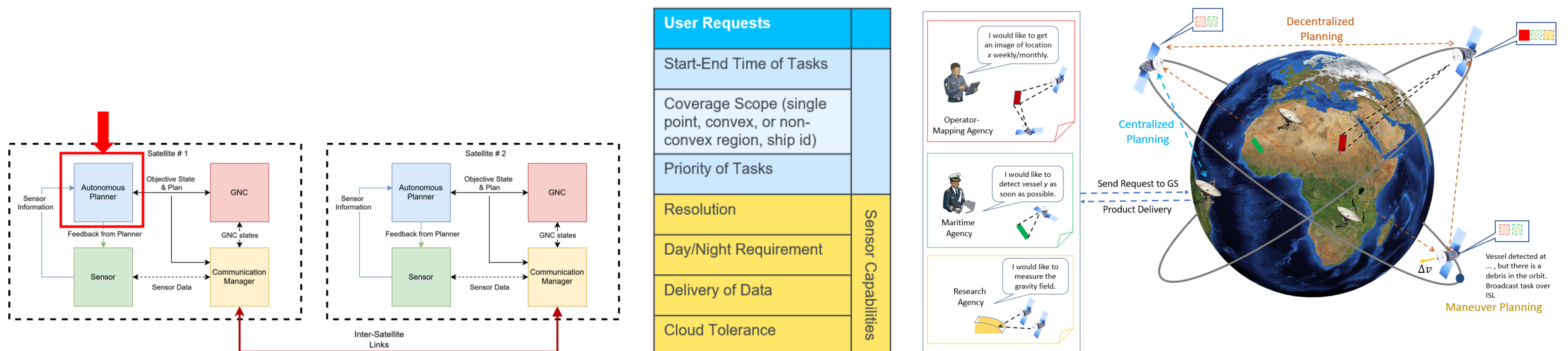
Motivation

- Multi-satellite missions are typically executed via direct ground commanding of each spacecraft, or through a central hub satellite that relays pre-computed plans to individual detector satellites.
- VLEO satellites has limited communication windows and transmission delays that makes centralized control impractical.
- Distributed space systems will take advantage of real-time decision making algorithms to manage mission planning and perform related maneuver planning without human intervention.
- Multi-mission + reconfiguration

Objectives

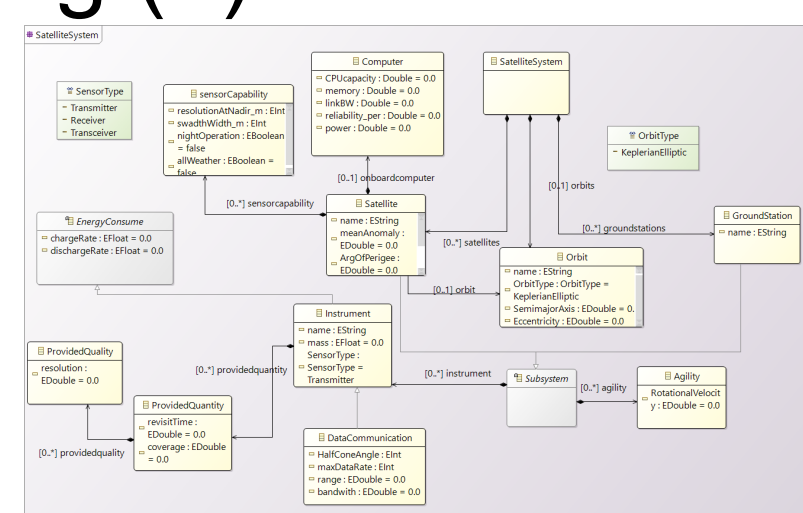
- Autonomous Decision Making: Developing algorithms for autonomous decision making in multi-satellite missions, enabling satellites to make decisions based on local information and interactions with neighboring satellites and ground stations.
- Decentralized Mission Planning: Creating decentralized mission planning algorithms that allow satellites to collaboratively plan and execute tasks without relying on a central controller.
- The scenario is that users request tasks (imagery tasks or data collection tasks) on-demand and receive the task as soon as possible.
- Reconfiguration(multi mission)- Maneuvering.

Research Highlights

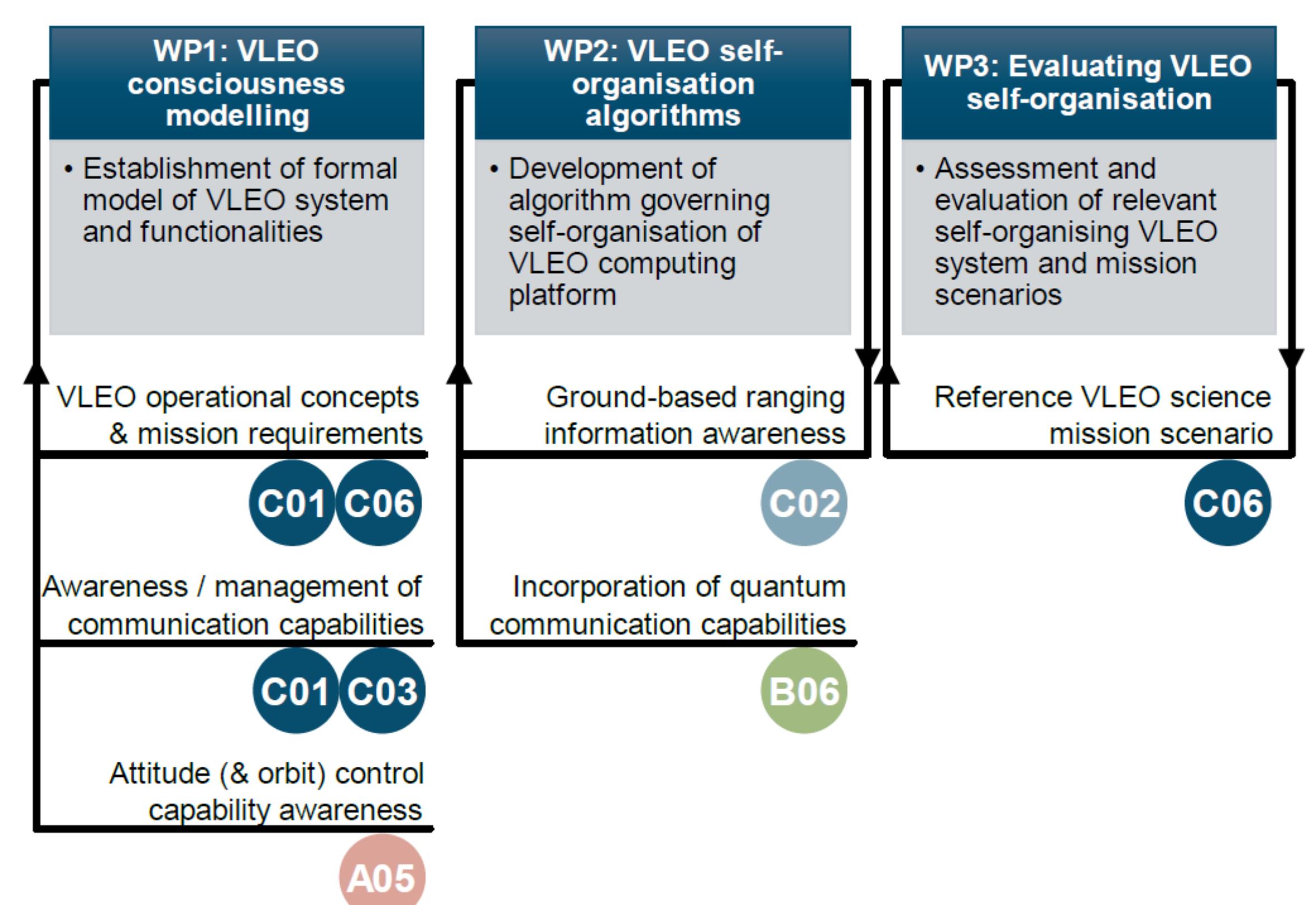


Methods

- Linear Programming
- Markov Decision Processes
- Reinforcement Learning (?)



Collaborations



References

Adams, C., Kempa, B., Iatauro, M., Frank, J., & Vaughan, W. (2023). An overview of distributed spacecraft autonomy at NASA Ames.

